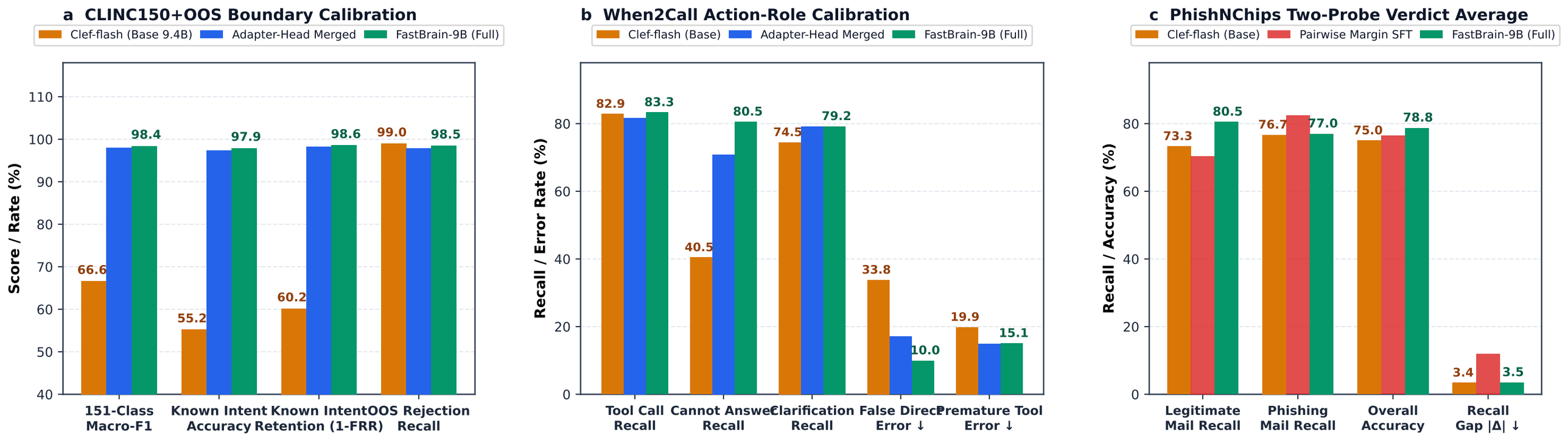
a CLINC150+OOS Boundary Calibration
Clef-flash (Base 9.4B)
Adapter-Head Merged
FastBrain-9B (Full)
Score / Rate (%)
110
100
90
80
70
60
50
40
66.6
98.4
98.4
55.2
97.9
97.9
60.2
98.6
98.6
99.0
98.0
98.5
151-Class Macro-F1
Known Intent Accuracy
Known Intent Retention (1-FRR)
OOS Rejection Recall
b When2Call Action-Role Calibration
Clef-flash (Base)
Adapter-Head Merged
FastBrain-9B (Full)
Recall / Error Rate (%)
80
60
40
20
0
82.9
83.3
40.5
80.5
74.5
79.2
33.8
17.0
10.0
19.9
15.1
Tool Call Recall
Cannot Answer Recall
Clarification Recall
False Direct Error ↓
Premature Tool Error ↓
c PhishNChips Two-Probe Verdict Average
Clef-flash (Base)
Pairwise Margin SFT
FastBrain-9B (Full)
Recall / Accuracy (%)
80
60
40
20
0
73.3
70.5
80.5
76.7
82.5
77.0
75.0
76.2
78.8
3.4
11.8
3.5
Legitimate Mail Recall
Phishing Mail Recall
Overall Accuracy
Recall Gap |Δ| ↓